

## Detailed Phase 1 Gantt Chart (Weeks 1 - 10)

■ = Active Learning/Building | ▨ = Practice/Review

| Week | Focus Area     | Detailed Topics                                                                                                  | Core Process to Understand (The "Why")                                                                                        | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 |
|------|----------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|----|
| W1   | Python & Git   | OOP, Data Structures (Lists/Dicts), Virtual Envs, Git (Commit/Push/Branch)                                       | <b>Process:</b> How to write modular code. PyTorch uses Object-Oriented Programming heavily. Git is mandatory for tech teams. | ■  | ▨  | ▨  | ▨  | ▨  | ▨  | ▨  | ▨  | ▨  |
| W2   | SQL & Linux    | Basic CLI ( <code>cd</code> , <code>ls</code> , <code>grep</code> ), SQL Joins, Group By, Window Functions, CTEs | <b>Process:</b> How to extract data from company databases. AI models need data; SQL is how you get it.                       | ▨  | ■  | ▨  | ▨  | ▨  | ▨  | ▨  | ▨  | ▨  |
| W3   | Linear Algebra | Vectors, Matrices, Dot Products, Matrix Multiplication                                                           | <b>Process:</b> AI models are just giant matrices multiplying each other. You need to know how data shapes change.            | ▨  | ▨  | ■  | ▨  | ▨  | ▨  | ▨  | ▨  | ▨  |
| W4   | Prob & Stats   | Mean/Variance, Normal Distribution, Probability, Bayes Theorem                                                   | <b>Process:</b> How to deal with uncertainty. ML is just statistics disguised as code.                                        | ▨  | ▨  | ▨  | ■  | ▨  | ▨  | ▨  | ▨  | ▨  |
| W5   | NumPy & Pandas | Arrays, Broadcasting, DataFrames, Missing Value handling, Filtering                                              | <b>Process:</b> How to clean dirty data. 80% of an ML Engineer's job is fixing bad data before training.                      | ▨  | ▨  | ▨  | ▨  | ■  | ▨  | ▨  | ▨  | ▨  |

[illegible]